INCH-POUND
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21 April 1992
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MILITARY SPECIFICATION

CLOTH, COATED (NYLON TAFFETA)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for chloroprene coated nylon cloth (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, Post Office Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1136	- Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

MILITARY

MIL-C-21852	- Cloth, Taffeta, Nylon

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487	- Glossary of Cloth Coating Imperfections
* MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2	- Packaging Requirement Codes

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580-0001).

* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

No. D-1424-83 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

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AMERICAN ASSOCIATION OF TEXTILE CHEMIISTS AND COLORISTS

No. 8-1988 - Colorfastness to Crocking: AATCC Crockmeter Method

(Copies may be obtained from AATCC National Headquarters, Post Office Box 12115, Research Triangle Park, NC, 27709-2215).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

* 3.3.1 Base cloth. The base cloth shall conform to type III of MIL-C-21852. The color shall match Green Shade 3406, piece number 6-60.

3.3.2 Coating compound. The coating compound shall be chloroprene rubber that is free from objectionable odor, plasticized and pigmented black. No elastomer other than chloroprene shall be used (see 4.4.1).

3.3.3 Coated cloth. The coating compound shall be applied to one side of the base cloth.

3.4 Physical requirements. The coated cloth shall conform to the requirements specified in Table I when tested as specified in 4.5.

Table I - Physical Requirements

Test	Requirements
Weight (Oz. per sq. yd.) (Grams per sq. meter)	4.7 (±0.5) (159.33)
Breaking strength (lbs) (N)	
Warp (Minimum)	110 (489.5)
Filling (Minimum)	95 (422.7)
Tearing strength (grams) (N)	
Warp (Minimum)	850 (8.33)
Filling (Minimum)	650 (6.37)

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Table I - Physical Requirements (cont.)

Test	Requirements
Hydrostatic resistance (lbs/sq in) (Pa) (Minimum)	
Initial	80 (5.5×10^5)
After 5 launderings 1/	30 (2.1×10^5)
After 3 dry cleanings 1/	60 (4.1×10^5)
After 40 hours weathering 1/	60 (4.1×10^5)
Flex stiffness (inch-lbs x 10^{-4}) (N m x 10^{-4}) (Maximum)	
At 70°F (44.6°C)	
Warp	2.5 ($.28 \times 10^{-4}$)
Filling	1.5 ($.17 \times 10^{-4}$)
At -40°F (-24.8°C)	
Warp	8.0 ($.90 \times 10^{-4}$)
Filling	4.5 ($.51 \times 10^{-4}$)
Adhesion of coating (lbs/2 inch width) (N/5.2cm) (Minimum)	10 (44.5)
Resistance to heat at 160°F (60.8°C)	No evidence of tackiness, exudation, or flexibility loss
Blocking (scale rating) (Maximum)	No. 2

1/ After subjection to laundering, dry cleaning, or weathering, but prior to hydrostatic testing, the specimens shall show no evidence of cracking, crazing, blistering, peeling, swelling, or tackiness.

3.5 Dusting medium. The coated cloth shall be dusted with a suitable dusting medium that does not support mildew growth (see 4.4.1). Only the minimum amount of dusting powder necessary to prevent blocking should be applied. The application of dusting powder shall be controlled during processing or shall be reduced by an after processing brush or wash so the coated side of the cloth shows not less than "good" resistance to crocking when tested as specified in 4.5.

3.6 Length and put up. The coated cloth shall be put up on rolls as specified in PPP-P-1136 (see 6.2). Alternately, the coated cloth may be put up on rolls in 120 to 160 yard lengths with no more than 4 pieces per roll and no piece less than 25 yards in length.

* 3.7 Width. The width of the coated fabric shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.8 Workmanship. The finished coated cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be visually inspected for compliance with color, finish, and appearance and shall be tested for the chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

* 4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition, the contractor shall furnish a certificate of compliance for the coating compound formulation requirement specified in 3.3.2 and for the non-mildew supporting characteristic of the dusting medium specified in 3.5.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2.

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4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be inspected on one side only. However, the side shall be alternated, so that alternate rolls are examined on the face and back respectively. The defects found shall be classified as listed below. In addition, the cloth shall be given a through-light inspection for pinholes. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted, regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. The lot size shall be expressed in units of one linear yard. The sample unit for this examination shall be one linear yard. The acceptable quality levels (AQL) for this examination shall be 1.5 for major defects and 4.0 for total defects (major and minor) per 100 units (yards). The inspection level shall be level II. The number of rolls from which the sample is to be selected shall be in accordance with Table II. An approximately equal number of yards shall be examined in each roll sampled.

Defect		Classification Major	Minor
I.	UNCOATED SIDE		
a.	Any hole, cut, tear, or smash	X	
b.	Any slub or multiple float		X
c.	Baggy, ridgy, or wavy cloth	X	
d.	Any strike through of coating		X
e.	Color shaded or streaked		X
f.	Any spot or stain		X
II.	COATED SIDE		
a.	Any cut, scratch, abraded area, hole or tear	X	
b.	Any uncoated area	X	
c.	Any area where coating is noticeably thinner	X	
d.	Any blister, tunnel or delamination of coating	X	
e.	Any lump or heavier coated area		X
f.	Foreign matter imbedded in coating	X	
g.	Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure	X	
h.	Coating cracked or checked	X	
i.	Coating not applied over full width of base cloth	X	
j.	Any pinhole	X	
k.	Any light area or window resulting from improper distribution of pigment		X
l.	Coating streak		X
m.	Any ripple, waviness, or dimensional distortion	X	
n.	Any spot, stain, or streak		X
o.	Width not as specified		X
p.	Color not as specified, shaded or streaked		X
q.	Any evidence of uncleanliness		X
r.	Any tackiness	X	
s.	Any objectionable odor (other than what is characteristic of the coating compound)		X

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Table II - Sample size

Lot Size (yards)	Sample size
Up to 1200 inclusive 1/	3
1201 to 3200 inclusive	5
3201 to 10,000 inclusive	8
10,001 to 35,000 inclusive	13
35,001 to 150,000 inclusive	20
150,001 and over	32

1/ If the lot contains fewer than 3 rolls, each piece in the lot shall be examined.

4.4.2.2 Examination for length.

4.4.2.2.1 Individual pieces. During the yard-by-yard examination of the cloth, each piece in the sample shall be examined for length. Any length found to be less than the minimum or more than the maximum specified, or more than two yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

* 4.4.3 Examination of packaging. An examination shall be made in accordance with the provisions of PPP-P-1136 to determine that put-up and preservation, packing, and marking comply with Section 5 requirements.

* 4.5 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilitzed in expressing the final results. The sample unit for test purposes shall be 4 continuous yards full width of cloth in the vulcanized state. The lot size shall be expressed in units of one yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

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Table III - Test methods

Characteristic	Requirement Reference	Test Method
Weight	Table I	5041
Breaking strength Warp and Filling	Table I	5100
Tearing strength Warp and Filling	Table I	ASTM-D 1424-83
Hydrostatic resistance Initial	Table I	5512 1/
After 5 launderings (cotton mobile)	Table I	5556 2/ and 5512 1/
After 3 dry cleanings	Table I	5508 2/ and 5512 1/
Weatherometer *(after 40 hours)*	Table I	5804 2/ 3/ and 5512 1/
Flex stiffness: At 70°F Warp and Filling	Table I	5206 4/
At -40°F Warp and Filling	Table I	5206 4/ 5/
Adhesion (lbs per 2 inch width)	Table I	5907
Resistance to heat	Table I	5850 6/
Blocking	Table I	5872
Resistance to crocking	3.5	AATCC 8-1988 7/

1/ Water pressure shall be applied against fabric side.
2/ After exposure, the specimens shall be examined for evidence of cracking, crazing, blistering, peeling, swelling, and tackiness.
3/ The coated side of the cloth shall be exposed to the carbon arc.
4/ The geometric mean of the coating up average and the coating down average shall be indicted. Five (5) determinations with coating up and five (5) determinations with coating down shall be made on each sample unit.
5/ The coated cloth shall be cooled for 1 hour at the specified temperature (± 2°F) prior to test.
6/ The cloth shall be exposed for 120 hours at 160°F and shall be examined for signs of tackiness, exudation or loss of flexibility when flexed or crumpled manually.
7/ The test shall be conducted using the blue crock cloth and shall be tested for resistance to dry crocking only.

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5. PACKAGING

* 5.1 Put-up and preservation. Put-up and preservation shall be level A or C (see 6.2).

5.1.1 Levels A or C. The coated fabric shall be put-up and preserved in accordance with the requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or C (see 6.2).

5.2.1 Levels A, B, or C. The coated fabric shall be packed in accordance with the requirements of PPP-P-1136, except that 4 rolls may be packed for any type of container specified.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with PPP-P-1136.

* 5.4 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coated fabrics covered by this specification are intended for use in the manufacture of special purpose cold weather clothing worn by Navy personnel.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, testing and approval of the first article.
d. Length required if other than specified (see 3.6).
e. Width required if other than specified (see 3.7).
f. Levels of preservation and packing required (see 5.1 and 5.2)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4).

6.3 Standard samples. For access to standard samples, address the procuring activity issuing the invitation for bids (see 3.1).

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6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Classification. Previously subject document had a Type I and Type II classification. Type II has been deleted. The remaining requirements are for the cloth previously identified as Type I.

* 6.6 Subject term (key word) listing.

Fabric, cold weather clothing
Fabric, special purpose clothing
Finish, chloroprene rubber

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

Review Activity:
DLA - CT

Project No. 8305-N399

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-19699E (NU)	92 04 21

3. DOCUMENT TITLE
CLOTH, COATED (NYLON TAFFETA)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

REVISION

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
CAULFIELD, LOUISE A.	Navy Clothing and Textile Research Facility	
c. ADDRESS (Include Zip Code) P.O. Box 50 Code 50 Natick, MA 01760-0001	d. TELEPHONE (Include Area Code) (1) Commercial (508) 651-4119 (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD) 92 04 21

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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